

重点在流程：HTTP 接收 -> 分词 -> 等待队列 -> 前缀匹配 -> Prefill/Decode -> 缓存写回 -> 流式输出

SGLang Online Inference Pipeline (v0.5.14)

1. Client HTTP

curl / OpenAI SDK POST /v1/chat/completions or POST /generate

2. FastAPI HTTP Server

python/sglang/srt/entrypoints/http_server.py receive JSON, create GenerateReqInput

3. TokenizerManager.generate_request

python/sglang/srt/managers/tokenizer_manager.py:576 tokenize text or accept input_ids

4. ZMQ send TokenizedGenerateReqInput

send_to_scheduler rid registered in rid_to_state wait on asyncio.Event

5. Scheduler.event_loop_normal

python/sglang/srt/managers/scheduler.py:1505 recv -> waiting_queue

6. get_new_batch_prefill + match_prefix_for_req

schedule_policy.py:85 RadixCache.match_prefix radix_cache.py:358

7. PrefillAdder / ScheduleBatch

allocate paged KV slots skip cached prefix extend_input_len = prompt - cached

8. Model worker Prefill then Decode

TpModelWorker forward RadixAttention kernel sample token decode loop

9. RadixCache write-back

cache_unfinished_req.py:485 during chunk cache_finished_req.py:438 on EOS

10. Detokenizer + streaming

BatchTokenIDOutput -> TokenizerManager.handle_loop:1824 SSE yield to client

11. Metrics

TTFT, TPOT, e2e latency, cached_tokens, prompt_tokens, completion_tokens